# Truth as a Trajectory: What Internal Representations Reveal About Large Language Model Reasoning

**Hamed Damirchi**[1] **Ignacio Meza De la Jara**[1] **Ehsan Abbasnejad**[2]
**Afshar Shamsi**[3] **Zhen Zhang**[1] **Javen Shi**[1]

[1]Australian Institute for Machine Learning, Adelaide University [2]Monash University
[3]Concordia University

[1]{firstname.lastname}@adelaide.edu.au [2]{firstname.lastname}@monash.edu
[3]afshar.shamsi@concordia.ca

arXiv:2603.01326v1 [cs.CL] 1 Mar 2026

## Abstract

Existing explainability methods for Large Language Models (LLMs) typically treat hidden states as static points in activation space, assuming that correct and incorrect inferences can be separated using representations from an individual layer. However, these activations are saturated with polysemantic features, leading to linear probes learning surface-level lexical patterns rather than underlying reasoning structures. We introduce Truth as a Trajectory (TaT), which models the transformer inference as an unfolded trajectory of iterative refinements, shifting analysis from static activations to layer-wise geometric displacement. By analyzing displacement of representations across layers, TaT uncovers geometric invariants that distinguish valid reasoning from spurious behavior. We evaluate TaT across dense and Mixture-of-Experts (MoE) architectures on benchmarks spanning commonsense reasoning, question answering, and toxicity detection. Without access to the activations themselves and using only changes in activations across layers, we show that TaT effectively mitigates reliance on static lexical confounds, outperforming conventional probing, and establishes trajectory analysis as a complementary perspective on LLM explainability.

## 1 Introduction

The deployment of Large Language Models (LLMs) in safety-critical domains, from legal reasoning to content moderation, has rendered the black-box evaluation of final outputs insufficient (Orgad et al., 2025; Shailya et al., 2025; Aljohani et al., 2025). While behavioral benchmarks measure what a model generates, they provide limited insight into how it arrives at its conclusions. Without visibility into the internal process, practitioners cannot reliably distinguish between a model that follows reasoning patterns correctly and one that merely relies on surface-level heuristics. This gap creates a fundamental bottleneck for safety. To trust these systems, we must be able to verify the validity of their internal thought processes, not just the probability of their tokens.

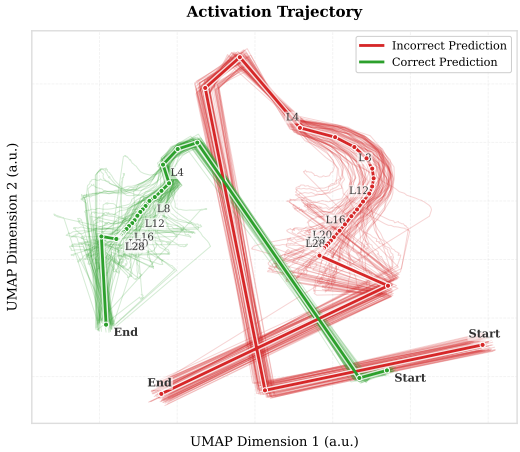


Figure 1: **Trajectories reveal structure beyond static embeddings.** We plot layerwise hidden states as trajectories in activation space. Correct generations (green) follow smoother paths, while incorrect ones (red) exhibit sharp deviations. Although the supervised projection amplifies separation, the geometry suggests that modeling entire trajectories, rather than isolated states, can help distinguish valid from spurious reasoning.

Recent mechanistic interpretability literature has been consistent with the Linear Representation Hypothesis (Park et al., 2023), which suggests that high-level properties such as inference validity or toxicity are encoded as distinct linear directions within the model's activation space. Thus, interpretability and behavior editing approaches assume certain properties can be classified via linear probes (Bao et al., 2025) or manipulated via activation steering (Aljohani et al., 2025; Rimsky et al., 2024; Anonymous, 2025; O'Brien et al., 2025) at carefully chosen layers. Under this view, explainability reduces to identifying the "right" static layer and direction that separates valid from invalid behavior. A limitation of these approaches is their reliance

on contrastive samples for the targeted property. Consequently, the efficacy of probing and steering results becomes subject to the specific datasets employed because they operate on static activations.

Thereby, despite early optimism, recent evidence suggests that the "geometries of truth" are often task-specific and orthogonal across domains (Azizian et al., 2025), i.e., a probe trained to detect inference correctness in one context fails to generalize beyond the distribution it was trained on. This failure may be driven by the polysemantic nature of transformer activations (Lindsey et al., 2025), which simultaneously encode lexical content, syntactic structure, and task-specific artifacts. This way, linear probes may latch onto surface-level correlations, such as the presence of specific tokens, rather than the underlying validity of the reasoning patterns. Furthermore, the selection of which layer to probe remains unprincipled. Findings from the activation steering literature indicate that effective intervention is only possible in a narrow mid-layer section of LLMs with inconsistent results across datasets (Rimsky et al., 2024), while other works suggest that the reasoning manifold is fundamentally non-linear (Manson, 2025), resisting simple linear separation. These inconsistencies are often a byproduct of evaluations on synthetic or tightly controlled datasets, leaving open the question of whether a generalizable, task-agnostic signature of reasoning validity exists in real-world benchmarks.

We propose **Truth as a Trajectory (TaT)**, which reframes LLM inference as a dynamical process rather than a collection of static layer snapshots. TaT unfolds the inference pass across *layers and tokens* into a trajectory through representation space (see Figure 1). We shift the analysis from raw activations to layer-wise displacements, i.e., the difference between successive residual stream activations across layers. This transformation mitigates reliance on static token-identity and lexical features, isolating how representations are updated across depth rather than what is encoded at a single layer. To capture the dynamics of this evolution, we use a lightweight LSTM classifier. While we explored simpler kinematic measures such as velocity, acceleration, and curvature inspired by recent work on transformer dynamics (Zhou et al., 2025; Fernando and Guitchounts, 2025), we found them to be inconsistent predictors of reasoning validity across diverse tasks. Our learned approach, on the other hand, captures the non-linear structural invariants associated with valid reasoning. To our knowledge, this is the first approach to model the "internal thought process" of an LLM on the reasoning validity of the input statement by unfolding the activations of every token across all layers into a continuous trajectory.

We evaluate TaT on widely used benchmarks spanning commonsense reasoning, question answering, factuality, and toxicity detection, across both dense and Mixture-of-Experts (MoE) architectures. Our main finding is that TaT achieves strong cross-dataset generalization. A trajectory classifier trained on a single source dataset generalizes across tasks with varying task-prompt structure, outperforming both linear probing baselines and the base model's own zero-shot and In-Context Learning (few-shot) performance on different domains. This suggests that the trajectory of valid reasoning encodes structural invariancies that transcend task-specific lexical patterns. On toxicity detection, TaT is more robust to lexical confounds, e.g., it can better distinguish quoted or contextualized toxic vocabulary from toxic intent and consistently outperforms probes trained on mid-layer or final-layer activations. Our contributions are summarized below:

- **Trajectory-based explainability**: We introduce Truth as a Trajectory (TaT), which models LLM inference as a dynamical process that unfolds across layers and tokens, capturing the continuous geometric evolution of reasoning rather than focusing on individual layers.
- **Cross-task Geometric Invariants**: By analyzing layer-wise displacement vectors rather than activations themselves, we mitigate reliance on static lexical features and emphasize the model's internal geometric refinement process, exposing trajectory-level structure that is unobservable to linear probes.
- **Trajectory-based behavior detection**: We demonstrate that trajectory analysis can be extended to complex behavioral properties, such as toxicity. TaT significantly outperforms linear probes in distinguishing meaningful context from toxic intent, validating its utility for reliable model monitoring.

## 2 Related Work

Mechanistic interpretability has traditionally focused on static linear representations of concepts within isolated layers. However, the residual stream

structure of Transformers suggests a dynamical systems perspective, where the evolution of representations across layers may provide insights into the thought process of the model. We situate our work at the intersection of these paradigms, focusing on the geometry of the inference trajectory.

### 2.1 Static Linear Representations

The current predominant paradigm in mechanistic interpretability is the *Linear Representation Hypothesis* (LRH), which suggests that neural networks represent high-level concepts as linear directions in activation space (Park et al., 2023; Elhage et al., 2022; Liu et al., 2025). This view has motivated works on classifying behaviors via linear probes (Belinkov, 2022) or interpreting them via Sparse Autoencoders (SAEs) (Huben et al., 2024). However, these methods face significant limitations. Both require exhaustive layer-wise searches, as the specific depth of concepts is unknown *a priori*, and SAEs notably lack consistent structural mapping across models, complicating cross-model interpretation. Furthermore, linear probing relies on subjective, curated datasets to define target behaviors, making the discovered directions dependent on specific semantic content rather than intrinsic model geometry. While extensions like Contrast-Consistent Search (CCS) (Burns et al., 2022) attempt to find latent knowledge without supervision they, along with standard probing, typically analyze representations as static vectors within isolated layers (e.g., "layer $L$"). This ignores the *temporal evolution* of the inference process. Our work challenges this static, content-dependent view, arguing that reasoning validity is a dynamic property best captured by the geometric displacement of activations across the computational trajectory, rather than their static positioning in fixed coordinates.

### 2.2 Representation Engineering and Steering

Representation Engineering (RepE) shifts focus from analysis to control. Zou et al. (2023) demonstrated that extracting "concept vectors" allows for top-down steering of model behavior, inhibiting hallucinations or toxic outputs by injecting these vectors into the residual stream. However, static steering is not guaranteed to predictably steer towards desired behavior for every model and task (Rimsky et al., 2024). This is potentially due to some datasets requiring the model's internal state to navigate a non-linear path. Thus, recent adjacent literature advocates in favor of alternative approaches. Zhang and Dong (2025); Manson (2025) argue that simple linear interventions cannot account for the "manifold evolution" during multi-step inference. Similarly, Postmus and Abreu (2024) suggests that steering should target activation regions via transformations rather than static directions. Meanwhile, like linear probing, these methods remain dependent on subjective, curated datasets to define target behaviors. Our approach mitigates these issues by analyzing the displacement of activations across the entire trajectory, capturing how activation trajectories behave in task-specific regions without relying on layer-wise activations.

### 2.3 Transformers as Dynamical Systems

The theoretical interpretation of Transformers as discretized dynamical systems provides the basis for our kinematic framework. The residual update $\mathbf{h}_{\ell+1} = \mathbf{h}_\ell + f(\mathbf{h}_\ell)$ is mathematically equivalent to a step of the Euler method for solving an Ordinary Differential Equation (ODE) (Chen et al., 2018; Lu et al., 2019). This equivalence implies that the layer-wise evolution of activations can be analyzed as a continuous trajectory in a high-dimensional state space, rather than as discrete, independent states. Geshkovski et al. (2023) extended this view to self-attention, modeling tokens as interacting particles that converge toward semantic clusters over "time" (depth). We conduct an extensive study on kinematic measures of activation trajectories across various models and datasets. We demonstrate that while pre-determined measures lack universal applicability, the intrinsic motion of internal representations may contain useful signals. Consequently, we model this motion through the activation space, showing that this trajectory can accurately classify the validity of the input statement's reasoning process or behavior characteristics.

### 2.4 The Geometry of Inference

Most relevant to our proposal is the emerging body of work on the "geometry of reasoning". Zhou et al. (2025) recently proposed that logical validity governs the "velocity field" of the representation flow, while semantic content determines position. They demonstrate that logical deductions trace specific flow patterns distinct from semantic association. However, their analysis remains largely theoretical, focusing on idealized logical structures rather than the noisy, unstructured data typical of real-world LLM usage. Concurrently, Manson (2025) introduced the concept of "Curved Inference," showing